

Module – 5

High-Voltage Electronics Fundamentals: High-Voltage DC Hazards, Safety of High-Voltage Electronics, Conductive Anodic Filaments, Floating Measurements, Y-Capacitance, HV Isolation, ESD Suppression on Isolated Devices, Isolation Detection.

Failure Detection: Overcharge/Overvoltage, Over-Temperature, Overcurrent, Battery Imbalance/Excessive Self-Discharge, Internal Short Circuit Detection, Detection of Lithium Plating, Venting Detection, Excessive Capacity Loss, Reaction Strategies.

High-Voltage DC Hazards

- A great deal of literature and institutional know-how exists about safety hazards for high-voltage ac systems found in nearly every commercial and industrial environment. With the advent of modern battery systems and power electronics, the existence of high-voltage dc systems is becoming increasingly common.
- Battery management system designers should be aware of important differences in safety aspects of ac and dc high-voltage systems. Interruption of dc currents in high-voltage systems is more challenging than ac currents. Because the current does not go to zero with each half-cycle as with ac, electric arcs created are not self-extinguishing.
- Relays and contactors for dc are specifically labeled and generally include features such as hermetically sealed enclosures filled with hydrogen and magnetic arc suppression/blowout. The dc disconnects are specially designed to break the dc arc when opened. Never use a contactor, disconnect or relay that is not rated for dc to interrupt dc.
- Electrolytic electromigration or “dendrite” growth, is the transport of conductive material across an insulating surface driven by an electric field. This phenomenon can cause unwanted shorts (as dendrites reach across insulating barriers and contact other conductive parts) or open circuits (as conductive material is removed from traces). The rate of growth is dependent on the applied voltage and the growth of dendrites is significantly more pronounced using high-voltage dc. Dendritic growth (see Figure 9.1) requires surface moisture and ionic contamination to initiate the process. High-voltage electronic PCB assemblies should therefore be manufactured, handled and installed with maximum care to prevent excess moisture and contamination

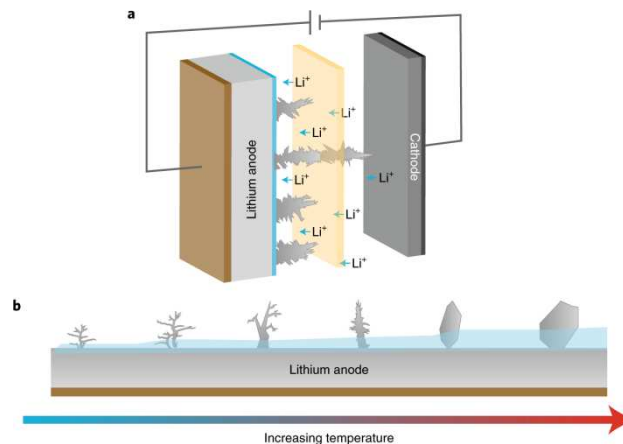


Figure 9.1 Dendrite growth

Safety of High-Voltage Electronics

- There are important differences in implementation between “normal” embedded control systems which generally operate with low-voltage control signals (between 5-V and 28-V dc) and high-voltage devices such as battery management systems for large-format lithium batteries.
- Most low-voltage embedded control systems have relatively low potential for electric arc hazards caused by dielectric breakdown or electric shock.
- High-voltage electronics must consider both of these potential risks. The potential for thermal events or fire exists with even low-voltage electronics and increases significantly as the maximum voltage encountered rises.
- The most obvious concern associated with high-voltage electronics is that of dielectric breakdown between two conductive parts at a large potential difference. This can occur within the context of a single printed circuit board, between traces, component leads, connector pins or mounting features. Break-down can also occur between PCBs, between PCBs and housings, between connector pins or between any two conductive parts—not all of which are necessarily intended to carry current.
- It is dangerous to simply apply the standard breakdown values for dielectric materials. Local field strengths around sharp edges may be much higher. Breakdown resistance is reduced with increasing humidity and contamination.
- The most appropriate way to prevent breakdown hazards is to ensure that the product design process ensures that appropriate creepage and clearance distances are maintained.
- Creepage is defined as the minimum path length between two conductive parts measured along the surface of a dielectric material. Creepage distances are required to prevent breakdown due to tracking, the creation of a conductive dendrite on the surface of an insulator.
- The following Figure 9.2 shows the difference between creepage and clearance distances between two conductive parts.

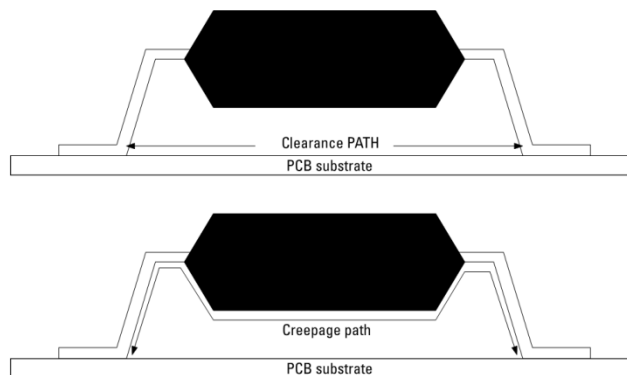


Figure 9.2 Creepage versus clearance distances

- Creepage distances can be increased by notching or slotting the PCB substrate. The comparative tracking index (CTI) of a material is a relative indicator of the susceptibility of the material to electrical breakdown along the material’s surface.
- The tests to establish the CTI of a given material measure the voltage required to cause tracking under standard test conditions. The CTI of the insulating material can affect the required creepage distance at a given working voltage.
- The prevalence of tracking is increased with the following factors:
 - High ambient humidity.
 - Presence of contaminants or corrosive materials.

- Low ambient pressure is a somewhat counterintuitive phenomenon that is a common condition with a high service altitude. Ground transportation may encounter altitudes up to 12,000 feet. Aviation may encounter altitudes of 50,000 feet or more. Aerospace applications may need to operate down to the vacuum of space.
- Clearance is defined as the minimum path length between two conductive parts measured through air. A number of standards exist that define the minimum creepage and clearance distances for a given working voltage and pollution degree.
- IEC 60950 and IEC 60664 are the most common of these standards. IEC 60950 and IEC 60664 provides minimum creepage and clearance values for a given material class, working voltage, pollution degree and insulation level. The pollution degree defines the characteristics of environment in which the device operates.
- The use of coatings can be used to improve the resistance to breakdown. Despite the high breakdown strength of conformal coatings, there are a number of areas in which care must be taken when using coated PCBs with high voltage.
- Coatings can be poorly or inconsistently applied with reduced effectiveness. Machine application of coatings is recommended for maximum uniformity. Inspection of coated boards can be performed under UV light to detect uncoated areas. Standards, such as IPC-A-610D, define acceptance levels for defects in coatings.
- The following best practices recommended for creepage and clearance safety on high-voltage circuit boards.
 - Refer to IEC 60950, IEC 60664, or a similar standard for creepage and clearance requirements.
 - Determine the appropriate pollution degree based on the installation in the application. A sealed enclosure may provide a lower pollution degree and therefore reduce the creepage and clearance distances.
 - Select the desired insulation type.
 - Determine the voltage levels and reference portals for individual elements of the system. Segregate areas of the circuit board according to voltage level and reference voltage.
 - Use an automated method wherever possible to determine the minimum distance between parts for evaluating creepage and clearance.

Conductive Anodic Filaments

- Conductive anodic filament (CAF) hazards are also a concern with high-voltage electronics. Conductive anodic filaments can form on printed circuit boards in humid conditions with high voltages between two plated through-hole features (vias or component pads) due to electromigration of copper ions from the PCB traces along the woven glass fiber of the PCB, producing a conductive filament which can cause an unintentional connection between two points at a large potential difference.
- The following figure shows the conductive anodic filament formation in PCB.

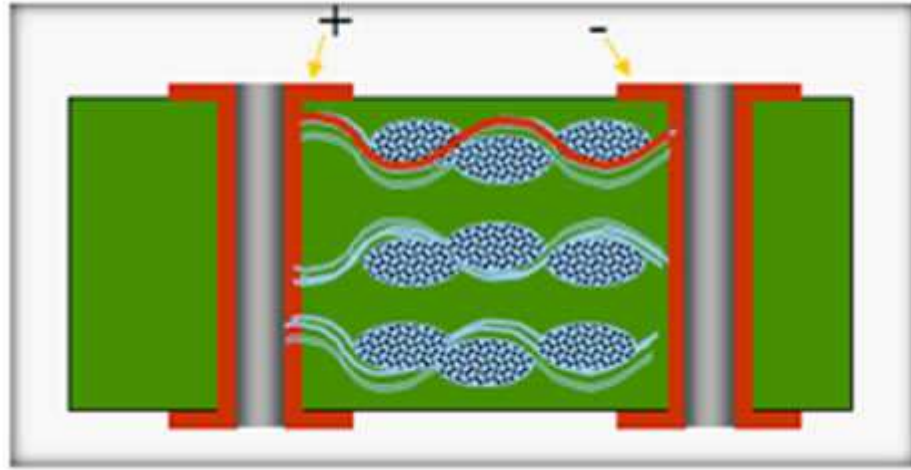


Figure 1: CAF formation between plated through hole

- To prevent CAF hazards from occurring, a number of practices should be followed in design and manufacturing of printed circuit boards exposed to high voltages.
- The copper filament conducts due to the presence of copper ions in a moist medium. Unlike dendrites which are metallic and conduct electronically, the CAF conductivity is ionic. Since the filaments form along the warp and fill directions, care can be taken to prevent aligning vias or through-holes with the glass fibers. Offsetting vias and holes will help avoid CAF formation if the glass fibers do not contact multiple vias at different potentials.
- The quality of the laminate also influences susceptibility to CAF hazards. CAF formation requires separation of the glass and epoxy which produces a path for ingress of copper ions and growth of the CAF. Absorption of water by the laminate can increase the susceptibility to this phenomenon.
- The manufacturing specifications for PCBs for high-voltage applications should clearly specify laminate materials which are appropriate for this application. IPC-4101B specifies requirements for base materials for PCBs, including quality levels and test methods.
- Unlike dendrites which grow on the surface of PCBs, CAFs are below the surface and cannot be readily seen. CAFs grow from an anode to a cathode, whereas dendrites grow from cathode to anode.
- IPC provides a test procedure (described in IPC-TM-650) that should be performed on PCBs to be used in high-voltage applications.

Floating Measurements

- There are important peculiarities about circuits designed to measure voltages that are not referenced to “earth” or “chassis” ground. In theory, the choice of any particular point in an isolated battery stack as a reference voltage against which to make cell measurements is as good as any other reference.

- However, in reality, many systems with isolated high-voltage components contain parasitic capacitances and conductances between isolated components and the earth/chassis ground.
- These parasitic elements introduce a number of complications to the design, analysis, and testing of large-format battery management systems.

Y-Capacitance

- The term Y-capacitance is used traditionally to describe capacitors placed intentionally between the ac input of a power supply and chassis ground as shown in the following figure 9.3).

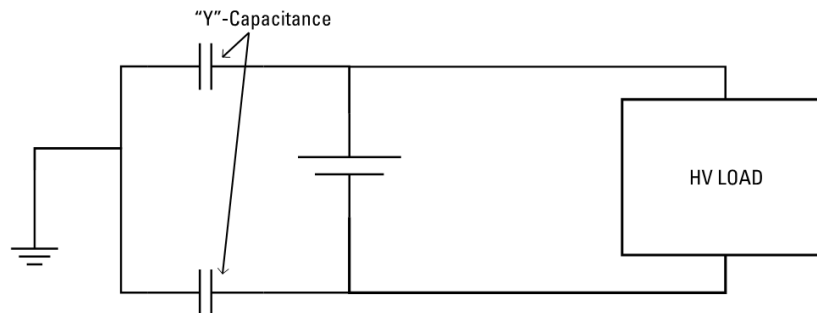


Figure 9.3 Battery system with Y-capacitance

- In the context of battery systems, Y-capacitance refers to capacitance, either intentional or parasitic, which exists between the high-voltage system and earth ground. While these capacitors are generally placed for reasons thought to be beneficial, they can create a number of problems. If they become too large, Y-capacitances pose a hazard to safety.
- Consider the following isolated battery system and its isolated load, with Y-capacitance installed between the two terminals of the load and earth ground.
- Let a fault is introduced by a person making unintended contact with part of the high-voltage system and the earth ground, as shown in the following Figure 9.4.

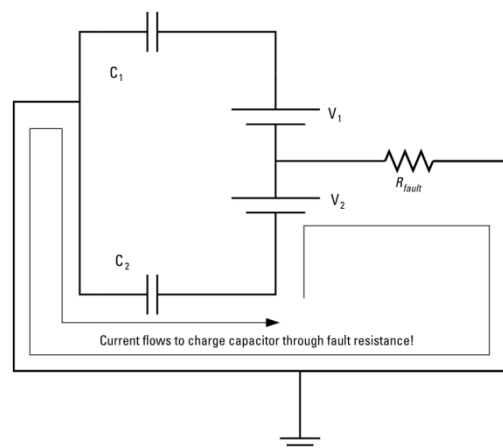


Figure 9.4 Fault in the presence of Y-capacitance

- Then the resistance of the fault is R_{fault} and considering the capacitor C_2 is uncharged, in a transient analysis, it can be replaced with a short circuit. The resultant circuit shows that the fault resistance will be connected through the discharged capacitor across a high potential. The energy that flows in this transfer is limited by the size of the Y-capacitance.

- Y-capacitances also pose a similar safety hazard when the system contains switching power electronics that generate ac voltages that can flow across capacitors that block dc voltages.
- Standards such as IEC 60479 detail the safety level that is associated with electric shock hazards of different magnitudes.

HV Isolation

- In many large-format systems, the dc battery voltage is not referenced to any earth or chassis ground. In the case of EV and HEV automotive battery systems, the high-voltage battery is not referenced to the 12-V chassis ground system at all.
- This provides an extra measure of safety in that a single point fault where the HV system is connected to earth ground does not create a large fault current. As such, electronic systems that are connected to the battery cells will need to provide electrical isolation barriers between the battery cells and earth ground.
- Power and signals will need to be transferred across these isolation barriers without compromising their isolation. On either side of an isolation barrier, the signals are referenced to different ground potentials that are not electrically connected. Because there is no fixed potential that can be used to measure the potential of these two systems, the potential difference between the two systems is arbitrary.
- It is often said that these two electrical systems are then “floating” with respect to one another, such that there may exist a significant potential difference between the reference (or “zero”) potentials of the two systems.
- Signals can be transmitted using different types of signal isolators. These can be separated into optical isolators (the signal is converted into light which is detected by another device on the opposite side of a transparent isolation barrier), capacitive isolators (a high-frequency signal is conducted across a small dielectric barrier that blocks dc), and magnetic isolators (electrical signals are converted to magnetic fields which cross a galvanic isolation barrier).
- Digital isolators are suitable for transferring high-speed digital signals such as RS-232, RS-485, SPI, I2C, or CAN communications across an isolation barrier.
- Each isolation circuit is generally capable only of transmitting a signal in a single fixed direction, but devices are available with multiple channels in various combinations of directionality to suit different communications strategies.
- High-speed digital isolators can create high radiated emissions if proper layout guidelines are not followed.
- Low-speed signals are usually isolated using optical isolation. The most common optoisolator is a light-emitting diode (LED) coupled with a photo-transistor in a common package. A dielectric insulating material is used between the two components. Because the switching side of the optoisolator uses a transistor device, the current on the switching

side can flow in only one direction. This is suitable for isolating signals but does not provide the bidirectional power flow required for many power connections.

- Optoisolators are available with different isolation ratings. There are usually two ratings available, the working voltage, which is the maximum voltage between the primary and secondary side during normal operation and the dielectric withstand voltage, which is the maximum allowable applied voltage (usually limited in time, frequently to 1 minute) between the two sides without risk of dielectric breakdown occurring.
- The working voltage rating should be used for the maximum voltage encountered in normal device operation and the withstand rating should be used for the maximum fault or abnormal condition that must be withstood. These ratings are based on both the creepage and clearance distances as well as the materials used for the optoisolator package.
- Optoisolated relays allow bidirectional current flow due to a complementary pair of transistors on the switching side. When the transistors are switched on, a bidirectional low-impedance connection exists, similar to a pair of contacts in a mechanical relay.
- Optoisolated relays are available in a variety of current ratings and voltage ratings. The relay will be rated for a maximum voltage blockage rating on the secondary side as well as a maximum voltage differential across the isolated barrier.
- Optoisolators and optoisolated relays used in a switching application will have a specified turn-on and turn-off time that in many cases will not be equal. All optically isolated components require careful analysis of the drive circuit. The LED must receive adequate drive current to ensure that the photo-transistor fully turns on.
- Optoisolated devices must be designed to provide adequate performance over the entire life of the device. Some optocouplers have an input driver IC which accounts for degradation of LED output.
- Optical isolators also introduce parasitic capacitance between the primary and secondary sides. This capacitance must be considered in the Y capacitance budget if a significant number of optical isolators are used.
- The following information is required when selecting any component designed to cross an isolation barrier:
 - Expected maximum working voltage across isolation barrier;
 - Expected maximum breakdown/overload/test voltage across isolation barrier
 - Required standards to be met by the devices;
 - Required creepage and clearance across isolation barrier.

ESD Suppression on Isolated Devices:

- Most electronic modules require testing to demonstrate they are capable of resisting electrostatic discharge (ESD), which can be created during use or during handling.
- ESD consists of an effective high-voltage (typically between 4 –25 kV) RC discharge into the circuit in question. If they are subjected to high voltages, even though the amount of energy in an ESD event is small, semiconductor devices, especially MOS transistors, can be permanently damaged.

- The voltage created during an ESD event, whether in service or during testing, is applied between the circuit and earth ground with either polarity.
- The standard method of preventing damage is through the use of capacitors, diodes or transient voltage surge suppressors (TVS) devices mounted close to connector pins that provide a low-impedance path for the ESD event to be conducted to ground. The circuit is usually mounted in some type of protective enclosure which prevents ESD from reaching sensitive components through other paths which are not protected in this way.
- If isolated circuits are included in the electronic device that is not referenced to ground as shown in the following Figure 9.5.

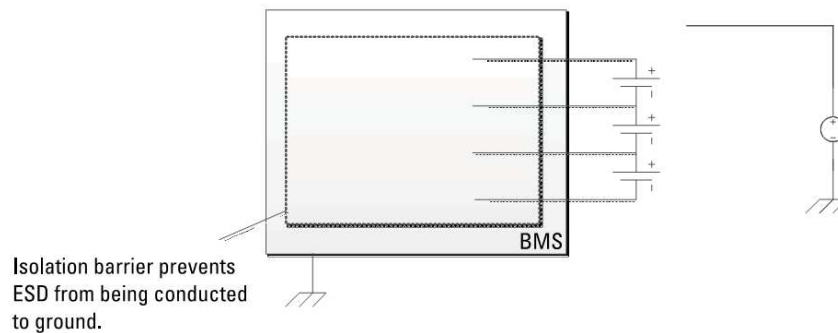


Figure 9.5 ESD path blocked by isolation barrier

- The energy from the ESD event will not be dissipated to ground. High voltages may be applied across isolation barriers, creating breakdowns, and circuits may easily be damaged. It is necessary to provide a path for ESD to reach earth ground and the use of high-voltage TVS devices offers one possible solution as shown in the following Figure 9.6.

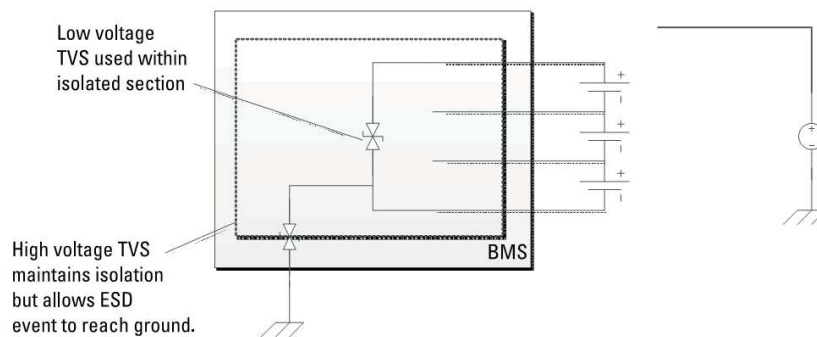


Figure 9.6 Possible ESD solutions

- TVS devices can be placed between the high-voltage inputs and earth ground. The TVS must have a breakdown voltage greater than the voltage expected on the inputs to ensure that the TVS does not conduct under normal operation, but this voltage must be low enough that the downstream components are properly protected. A cascading combination of TVS and capacitors may allow a reduction in the number of TVS required.
- High-voltage capacitors are another option. These represent a significantly higher cost than low-voltage capacitors typically used for ESD suppression. These capacitors need to be installed between the high-voltage and low-voltage systems and therefore effectively span the isolation barrier, therefore increasing the system Y-capacitance.

Isolation Detection

- On systems with isolated battery stacks, detection of an isolation fault or ground fault (an unintentional current path between the battery stack and components at or referenced to earth ground potential) is an important feature. Because the remainder of the system may not be energized if the battery is disconnected, it makes sense to place this functionality inside the battery system.
- The method most commonly used to detect isolation faults is the method used for static measurements of isolation in accordance with the Federal Motor Vehicle Safety Standard (FMVSS) 305 test performed on battery-powered vehicles.
- Let a fault of resistance R_{fault} exists between the battery system and earth ground as shown in the following figure 9.7. This fault can occur at any potential of the battery system.

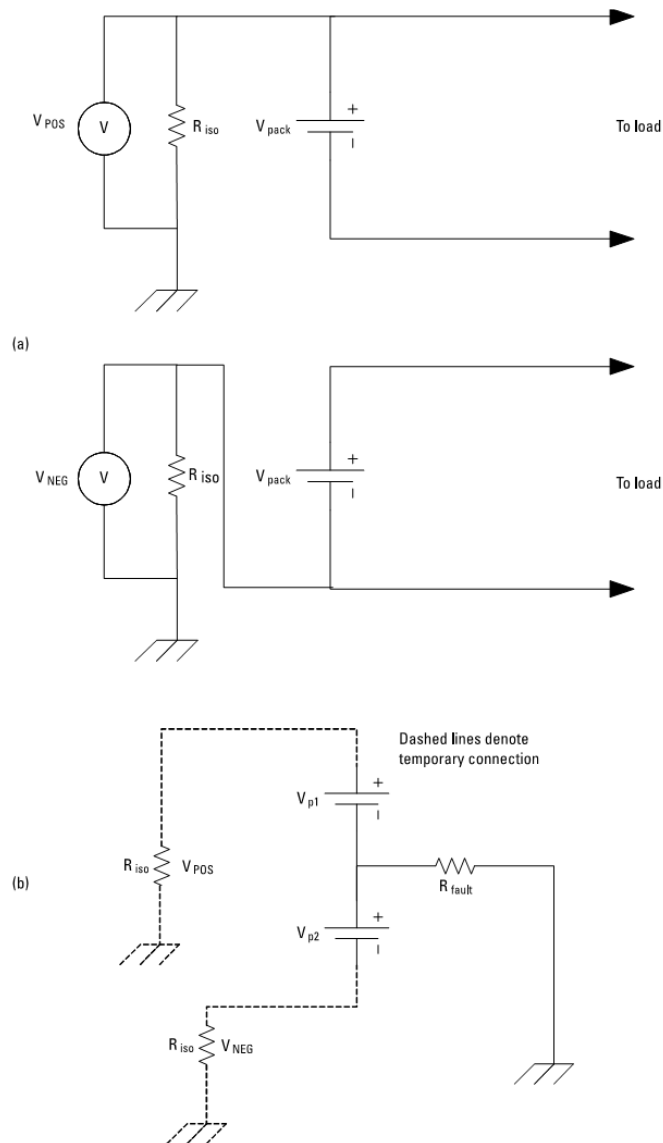


Figure 9.7 Isolation resistance measurement

- Assuming the general case where the fault occurs somewhere between the two terminals of the battery, effectively dividing the battery pack into two subpacks and creating the following equivalent circuit.
- The value of R_{fault} can be calculated by making two measurements. For each measurement, a known resistor R_{iso} is inserted between the positive or negative terminal of the battery and earth ground. The voltage across this resistor is measured.
- This isolation measurement can be made periodically. Continuous isolation measurement is usually desirable to provide constant monitoring of this important safety parameter. However, it is important to understand the timing limitations of this circuit.
- Consider the same circuit, but include the presence of Y-capacitance between the high voltage bus and earth ground. When the measurement resistor R_{iso} is inserted, it now forms an RC circuit using the Y-capacitance. The capacitor must fully charge before the measurement can be made. This establishes a minimum time required for making this measurement and therefore a maximum frequency at which it can be made.
- Isolation faults can occur due to the ingress of moisture into battery or electronics systems, due to the presence of foreign objects or materials (conductive debris) that bridge gaps between conductive parts, due to the formation of conductive bridges on circuit boards, faults internal to isolating components, insulation breakdown, and other hazards.
- Isolation faults are hazardous to those working on battery systems and multiple distributed isolation faults at different potentials can create short-circuit hazards. Therefore, many applications specify that operation with compromised isolation resistance is prohibited or must be accompanied by a warning to users.

Fault Detection

- The battery management system is required to detect defects in the battery cells as well as the balance of plant to ensure that a damaged battery that is susceptible to becoming hazardous with further use is not placed in service.
- The detection mechanisms must be accurate but robust against nuisance errors which reduce battery reliability and availability. Reactions to battery failures must be timely and appropriate and coordinated between the battery management system and the load devices to preserve safety.
- Failure rates for lithium-ion cells depend on the maturity of the manufacturing processes for the particular cell. The 18650 cylindrical cells are manufactured in extremely high numbers and failure rates well below one internal defect per million cells are achieved

by most reputable manufacturers. However, when these cells are used in large arrays, the failure rates at the system level may become appreciable again.

- Other cell technologies may be significantly less mature and have a much higher probability of a cell defect.

Failure Detection

Overcharge/Overvoltage

- Detection of overcharge or over discharge is crucial to avoiding the multiple hazardous effects that these conditions create. The first line of defense is accurate and complete cell voltage measurements.
- Faults should be set at the edge of the normal operating range and the system should respond rapidly by reducing limits and disconnecting the battery if overcharging or over discharging is occurring.
- A second layer of protection can be incorporated by measuring the rate of change of voltage as shown in the following Figure 17.1.

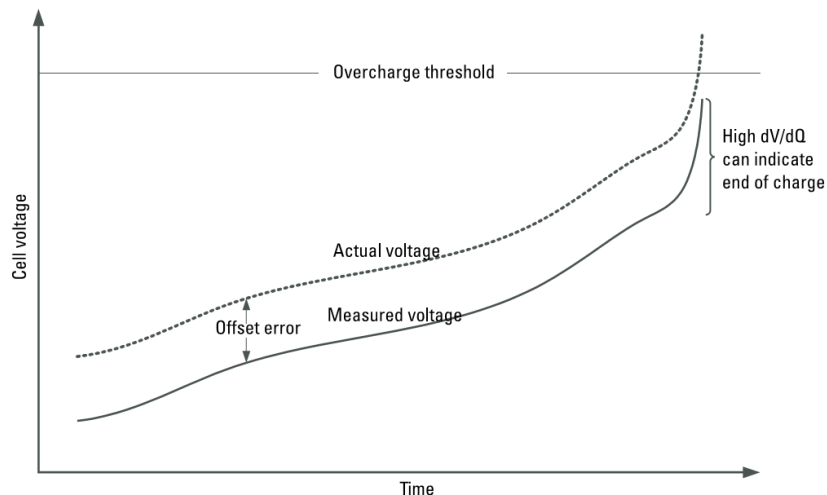


Figure 17.1 End-of-charge voltage error detection scheme.

- As most cells approach overcharge or overdischarge, the cell voltages begin to rise or fall much more rapidly than in normal operation. Bounds can be set on the maximum expected rate of voltage change with respect to cell current. The slope of the OCV(SOC) curve is the rate of change of voltage with respect to rate of change of state of charge.
- Maximum expected rate of change can be established for the IR-compensated voltage as a function of the SOC and current. If these rates of change are exceeded, it is possible that the battery is experiencing one of the following hazardous conditions.
 - The cell is exceeding the safe voltage range and entering overcharge or overdischarge, but a voltage offset measurement error is causing the cell measurement to appear safe.

- The cell is approaching the edge of the safe operating area. For systems with high power-energy ratios that operate over wide SOC windows, this technique can be used to allow higher charge and discharge rates without risk of overcharge, overdischarge, or cell reversal. This could be due to errors in SOC estimation.
- For battery systems that are charged at high rates to high states of charge, overcharge may happen very quickly. Most lithium-ion cells will display a rapid increase in voltage once the cell exceeds 100% SOC.
- An effective means of preventing sudden overcharge, if the battery charger can operate with a volt-age limit, is to continually command the charger to enforce a pack voltage limit which is only slightly higher than the current battery pack voltage, and to continually update this limit, “floating” it just above the measured pack volt-age. If an individual cell voltage rises suddenly, the charger will encounter the limit voltage and stop charging.

Over-Temperature

- In the same way as voltage rates of change are limited, the heat capacity and expected heat generation will limit the rate at which temperatures in a large-format battery system can change. However, in the event of a thermal event, the amount of available energy will be much higher leading to rapid temperature rises.
- Rapid decreases in measured temperature that exceed the physical limitations of the cooling system (if there is one installed) can safely be disregarded as measurement errors.
- Because over-temperature conditions usually lead to shutdown of the battery system, avoiding nuisance trips of over-temperature faults is critical.
- Therefore, a possibility check on high-temperature events may be needed to improve system reliability. This must be weighed against the potential hazard of delaying battery shutdown in the event of an over-temperature event.

Over current

- Very high currents should be prevented by a passive overcurrent protection device (generally a fuse or a circuit breaker) that requires no input from the battery management system to operate.
- The detection of a blown fuse is usually an important feature and can be accomplished by measuring high voltage on both sides of the fuse. Fuses react generally very quickly (especially if they are of the semiconductor type designed to protect power electronics), but all fuses clear more slowly with lower current, and larger fuses therefore allow more energy and higher current pulses before opening.
- Oversizing the fuse can result in other battery system components being exposed to potential over current. In this case the system may rely upon the battery management

system to prevent moderate over currents of sufficient duration to create a hazard but not long enough to clear the fuse.

- The current sensor must have adequate measurement range to measure the applicable range of over currents. The increased measurement range will reduce the accuracy and precision of the current measurement during normal operation.
- The measurement system must also consider the possibility that sensor saturation occurs due to a sensor malfunction. Excessive discharge currents will be accompanied with significant voltage depression for all cells that can be used as a plausibility check for the high current.
- The battery management system can command contactors to open to interrupt excessive currents. If contactors are to be opened under high current, they must be rated to interrupt the currents expected.

Battery Imbalance/Excessive Self-Discharge

- If the battery cells begin to experience an excessive rate of self-discharge, the balance of the battery will become negatively affected if the differences in self-discharge cannot be compensated by the cell balancing system.
- In many cases excessive self-discharge will affect only a small number of cells due to a defect. If the SOC and capacity for all individual cells are being computed, then determining imbalance and self-discharge is a relatively straightforward process.
- In the more common scenario in which the SOC and capacity information is calculated on an aggregate basis, other methods may be used to determine imbalance. At a given aggregate SOC, the SOC for each individual cell should be expected to fall within a narrow range of $SOC_a - \varepsilon$ to $SOC_a + \varepsilon$.
- The slope of the SOC-OCV relationship is known throughout the SOC range, which can be used to determine a maximum expected voltage difference between cells. This approach can be used with care during battery operation but becomes more robust if the voltage measurements can be made while the battery current is zero (to avoid synchronization issues) and has been so for some time (to avoid differences in internal cell resistances and dynamics to influence the result).
- A relatively simple check for self-discharge can be made even if the individual SOC of all cells is not accurately known simply using the relative rates of voltage decrease. If cells are close in SOC, then equal losses of charge should result in approximately equal decreases in voltage.
- If the battery system experiences long periods of inactivity where these can be measured (for example, in an electric vehicle that is turned off for many hours per day, or a backup power system that normally does not provide any current if the primary source of electricity is operational), the rate of voltage decrease can be simply calculated even with a single sample per day.
- Because all cells have seen the same current excitation, the internal dynamics should relax at the same rate. If a particular cell exhibits an excessively high rate of self-discharge, the voltage decrease should be evident after a few days of sampling if voltage measurements are

accurate—appropriate low pass filtering is needed, but this method avoids the complexity of calculating all individual cell SOC and requires little sampling.

Internal Short Circuit Detection

- This is a challenging area of active research. Many methods focus on the detection of a change in the shape and/or duration of the charging taper or in detecting noisy voltage signals which occur as the cell approaches 100% SOC at low current.

Detection of Lithium Plating

- This is a relatively new concept that has been discussed in academic literature [2]. If metallic lithium is deposited in a lithium-ion battery due to abuse, due to the lower anode voltage created by the lithium anode, the shape of the OCV/SOC relationship will change on discharge with a higher-voltage plateau appearing at the start of a discharge cycle.
- Low discharge currents, which could be possibly created by the balancing circuit, will show a higher cell voltage than cells without lithium plating.
- Significant developments are needed to ensure this concept is robust, but detecting this condition is desirable, and likely this will become a feature in many large-format battery management systems in the future.

Venting Detection:

- Recently, sensors have become available designed to detect volatile organic compounds in electrolyte. The presence of volatile organic compounds in a battery system is often associated with cell venting and may provide an early warning of this hazardous event.
- These sensors can detect cell venting regardless of the cause (over-temperature, manufacturing defect, mechanical intrusion, overcharge) and provide improved safety in the event of cell venting.
- These technologies are still relatively new and should not be relied upon for primary protection.

Excessive Capacity Loss

- An important distinction should exist between normal loss of capacity due to aging phenomena and excessive, rapid loss of capacity due to a battery defect.
- An effective capacity detection scheme can be coupled with an expected worst-case capacity trajectory to determine if the battery is declining more quickly than usual. Often capacity loss will be due to unmanageable self-discharge in one or more cells.
- Significant and unexpected capacity loss may indicate a serious internal defect due to internal shorts, cell damage, or lithium plating.

Reaction Strategies

- When a fault occurs, the battery management system should bring the battery to a safe state as quickly as possible, while maintaining an appropriate level of functionality.
- The definition of all of these reactions is specific to the nature of the battery system as well as the specific type of error condition being addressed.
- Fault strategies should always take a secondary role in protection if primary protection methods, such as limit algorithms fail to protect the battery from hazardous conditions.
- The various systems of the battery management system should be designed such that if the load respects the limits published by the battery and the battery SOH is acceptable, fault or error conditions should not occur.
- Faults should be associated with a system malfunction, unexpected battery response, or degradation or unexpected load condition.
- A common strategy for serious faults that do not require immediate disconnection of the battery, but need a timely response, is to allow limited operation for a specified period of time, but to prevent restarts or reconnection of the battery system after this period has passed. This can allow adequate energy for the load device to execute an orderly shutdown.
- In the event of impending overcharge or over discharge in which the load has failed to respond to requests to reduce battery current to zero, the battery can simply be disconnected from the load to prevent further changes in battery SOC. Although this will prevent further charging or discharging, damage to battery system components may result.
- Stationary energy storage systems often incorporate fire suppression systems that could be activated by the battery management system. The operation of any last-resort safety systems such as this should have appropriate measures such as requiring at a minimum, no single-point failures in hardware or software will lead to inadvertent actuation.